SMART LIGHT MIRROR

Design Thinking for Innovation & Entrepreneurship (DTIE)

Section: 1 | Project Batch: A | Group 1B
Arduino Uno • HC-SR04 Ultrasonic Sensor • LED Lighting

Team Members

Project: Smart Light Mirror

Group 1B | Subject: DTIE Projects

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Vision and Mission

VISION

Make everyday mirror use more convenient through simple, automatic lighting.

MISSION

Build a low-cost prototype that detects a person near the mirror and switches the LEDs automatically.

DESIGN PRINCIPLES

Automatic operation • Quick response • Simple setup • Practical use of readily available components

Problem Statement

Conventional mirror lighting often depends on manual ON/OFF switching.

Poor lighting can make mirror use difficult, while reaching for a switch can be inconvenient.

Lights may also be left ON when they are no longer needed.

DESIGN QUESTION

How can mirror lighting turn ON when a person approaches and turn OFF when they move away?

Research / Facts / Market & Industry Study

OBSERVATIONS FROM OUR PROBLEM STUDY

- Manual switching adds an extra step to a daily routine.
- Users need suitable illumination while standing near a mirror.
- Automatic proximity-based control is one possible way to reduce unnecessary lighting.

TECHNOLOGY REVIEW

We considered manual switches, IR, LDR, PIR and ultrasonic sensing. The HC-SR04 was selected because it measures distance without physical contact.

Note: The source presentation contains no verified market-size figures or survey statistics; this slide presents qualitative findings only.

Target User - Customer

PRIMARY USERS

- Students
- Home users
- Working professionals
- People using mirrors in low-light conditions

USER NEEDS

Convenient operation • Better visibility • Quick response • Simple controls • Avoiding lights left ON unnecessarily

Competitors / Benchmarks

ALTERNATIVES CONSIDERED

Manual switch — simple, but requires manual operation.

IR sensor — can detect nearby objects; considered as an alternative.

LDR — responds to light level rather than a person's distance.

PIR sensor — detects movement; considered as an alternative.

HC-SR04 ultrasonic sensor — provides distance-based detection and was selected for this prototype.

SELECTION BASIS

Cost, detection capability, programming ease, component availability, response and prototype feasibility.

Risks and Gap

POTENTIAL RISKS

- Sensor readings may vary with distance, angle or the surrounding environment.
- Incorrect distance thresholds may switch the LEDs at the wrong time.
- Loose breadboard or jumper-wire connections may affect operation.
- Power and wiring must be checked before testing.

CURRENT GAP

The prototype demonstrates proximity-based lighting; it does not provide interactive mirror features or advanced smart-home functions.

MITIGATION

Test at different distances, adjust the programmed threshold, secure connections and repeat the ON/OFF tests.

Concept / Idea / Prototype / Product / Solution

SOLUTION

An Arduino-based Smart Light Mirror that automatically controls LED lighting based on how close a person is to the mirror.

HOW IT WORKS

1. HC-SR04 sends an ultrasonic signal and receives its echo.
2. Arduino Uno calculates the distance.
3. The C++ program checks the distance condition.
4. LEDs turn ON when a person enters the detection range and turn OFF when the person moves away.

CORE COMPONENTS

Arduino Uno • HC-SR04 ultrasonic sensor • LEDs • Breadboard • Jumper wires • Mirror • USB/power connection

Images / Storyboard / Architecture

SYSTEM ARCHITECTURE

PERSON APPROACHES MIRROR

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HC-SR04 MEASURES DISTANCE

↓

ARDUINO UNO PROCESSES READING (C++ / ARDUINO IDE)

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DISTANCE CONDITION CHECKED

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LED LIGHTS TURN ON / OFF

CONNECTION OVERVIEW

HC-SR04 VCC → 5V; GND → GND; TRIG and ECHO → Arduino digital pins. LED connects to an Arduino output through the breadboard; jumper wires complete the circuit.

Lean Canvas Model

PROBLEM

Manual mirror-light switching; inconvenient use; lights can remain ON unnecessarily.

SOLUTION

Proximity-based LED control using HC-SR04 + Arduino Uno.

CUSTOMER SEGMENTS

Students, home users, working professionals and users in low-light conditions.

UNIQUE VALUE

Simple, hands-free mirror lighting using commonly available components.

CHANNELS / KEY METRICS

Prototype demonstration; test successful detection and correct LED ON/OFF response at different distances.

COST / ADVANTAGE

Uses a small set of standard electronics components; intended as a low-cost educational prototype.

Capital (Financial)

MAIN COST ITEMS

- Arduino Uno
- HC-SR04 ultrasonic sensor
- LEDs
- Breadboard and jumper wires
- Mirror and USB/power connection

FINANCIAL APPROACH

The design uses readily available components and is intended to remain low-cost.

The source presentation does not provide purchase prices or a total budget, so exact costs should be added after checking the components actually purchased.

Roadmap / Timeline

1. IDENTIFY THE NEED

List the difficulties with manual mirror lighting.

2. EXPLORE IDEAS

Compare manual, IR, LDR, PIR and ultrasonic approaches.

3. SELECT THE CONCEPT

Choose HC-SR04 with Arduino Uno based on cost, distance detection and prototype feasibility.

4. BUILD THE PROTOTYPE

Connect the sensor, Arduino and LEDs using a breadboard and jumper wires.

5. PROGRAM AND TEST

Write C++ code in Arduino IDE; test person-near/person-far conditions and adjust the distance threshold.

6. DEMONSTRATE AND IMPROVE

Show LED ON/OFF behaviour, check connections and refine reliability based on testing.

Conclusion

- **The Smart Light Mirror prototype applies the robotics cycle: SENSE → PROCESS → ACT.**

The HC-SR04 detects distance, the Arduino Uno processes the reading, and the LEDs respond to the programmed condition.

Thank You | Group 1B